

DevOps Capstone Project 1

Retail E-Commerce Order Management Platform

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1. Technologies Used:

- Trello
- GitHub
- GitHub Actions
- Python Flask
- Pytest

2. Project Overview

This project demonstrates a DevOps-based CI workflow for a Retail Order Management Platform using Agile planning, source control management, automated testing, and CI pipeline automation.

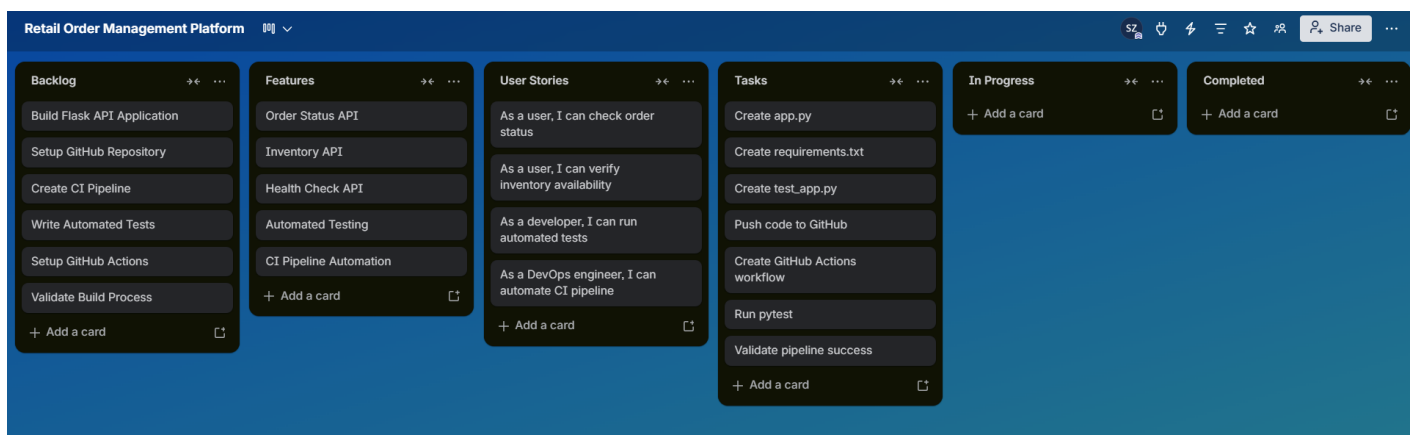
The application provides APIs for:

- Health Check
- Order Status
- Inventory Status

The project uses GitHub Actions to automatically run tests whenever code is pushed to the repository.

3. Agile Planning using Trello

A Trello board was created to manage Agile planning activities including Features, User Stories, Tasks, and workflow tracking.



4. Flask Application Development

A Flask-based Python application was developed to expose APIs for:

- Home Page
- Health Check
- Order Status
- Inventory Status



Retail Order Management Platform



Order Service Healthy



Order Processing Successfully



Inventory Available

5. Application Code

The application was developed using Python Flask framework and structured using modular files including:

- app.py
- requirements.txt
- test_app.py

6. Testing using Pytest

Automated testing was implemented using Pytest to validate all API endpoints and ensure successful responses.

```

PS C:\Users\zerub\OneDrive\Desktop\retail-order-management-platform> python -m pytest
===== test session starts =====
platform win32 -- Python 3.11.9, pytest-9.0.3, pluggy-1.6.0
rootdir: C:\Users\zerub\OneDrive\Desktop\retail-order-management-platform
collected 4 items

test_app.py .... [100%]

===== 4 passed in 3.57s =====

```

7. Source Control using GitHub

The project source code was pushed to GitHub repository for version control and collaboration.

```

PS C:\Users\zerub\OneDrive\Desktop\retail-order-management-platform> git init
Initialized empty Git repository in C:/Users/zerub/OneDrive/Desktop/retail-order-management-platform/.git/
PS C:\Users\zerub\OneDrive\Desktop\retail-order-management-platform> git add .
PS C:\Users\zerub\OneDrive\Desktop\retail-order-management-platform> git commit -m "Initial retail order management application"
[master (root-commit) 0411f9e] Initial retail order management application
6 files changed, 45 insertions(+)
create mode 100644 .github/workflows/python-app.yml
create mode 100644 __pycache__/app.cpython-311.pyc
create mode 100644 __pycache__/test_app.cpython-311-pytest-9.0.3.pyc
create mode 100644 app.py
create mode 100644 requirements.txt
create mode 100644 test_app.py
PS C:\Users\zerub\OneDrive\Desktop\retail-order-management-platform> git remote add origin https://github.com/ShinyZerubbaRajiveS/retail-order-management-platform.git
PS C:\Users\zerub\OneDrive\Desktop\retail-order-management-platform> git branch -M main
PS C:\Users\zerub\OneDrive\Desktop\retail-order-management-platform> git push -u origin main
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/ShinyZerubbaRajiveS/retail-order-management-platform.git/'
PS C:\Users\zerub\OneDrive\Desktop\retail-order-management-platform> git config --global credential.helper manager
PS C:\Users\zerub\OneDrive\Desktop\retail-order-management-platform> git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 12 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (11/11), 2.55 KiB | 201.00 KiB/s, done.
Total 11 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/ShinyZerubbaRajiveS/retail-order-management-platform.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\zerub\OneDrive\Desktop\retail-order-management-platform> git rm -r --cached __pycache__
rm '__pycache__/app.cpython-311.pyc'
rm '__pycache__/test_app.cpython-311-pytest-9.0.3.pyc'
PS C:\Users\zerub\OneDrive\Desktop\retail-order-management-platform> git add .
warning: in the working copy of '.pytest_cache/CACHEDIR.TAG', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\zerub\OneDrive\Desktop\retail-order-management-platform> git commit -m "Added gitignore and removed pycache"
[main 350d751] Added gitignore and removed pycache
4 files changed, 20 insertions(+)
create mode 100644 .pytest_cache/.gitignore
create mode 100644 .pytest_cache/CACHEDIR.TAG
create mode 100644 .pytest_cache/README.md
create mode 100644 .pytest_cache/v/cache/nodeids
PS C:\Users\zerub\OneDrive\Desktop\retail-order-management-platform> git push
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 12 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (9/9), 985 bytes | 164.00 KiB/s, done.
Total 9 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/ShinyZerubbaRajiveS/retail-order-management-platform.git
0411f9e..350d751 main -> main

```



retail-order-management-platform Public



Pin



Watch 0



main



1 Branch



0 Tags



Go to file



Add file



Code



ShinyZerubbaRajiveS Added gitignore and removed pycache

350d751 · 2 minutes ago

2 Commits



.github/workflows

Initial retail order management application

7 minutes ago



.pytest_cache

Added gitignore and removed pycache

2 minutes ago



__pycache__

Initial retail order management application

7 minutes ago



app.py

Initial retail order management application

7 minutes ago



requirements.txt

Initial retail order management application

7 minutes ago



test_app.py

Initial retail order management application

7 minutes ago

8. CI Pipeline using GitHub Actions

GitHub Actions workflow was created to automate:

- Dependency installation
- Build validation
- Automated testing

The workflow automatically triggers whenever code is pushed to the main branch.

9. GitHub Actions Workflow File

Workflow File:

.github/workflows/python-app.yml

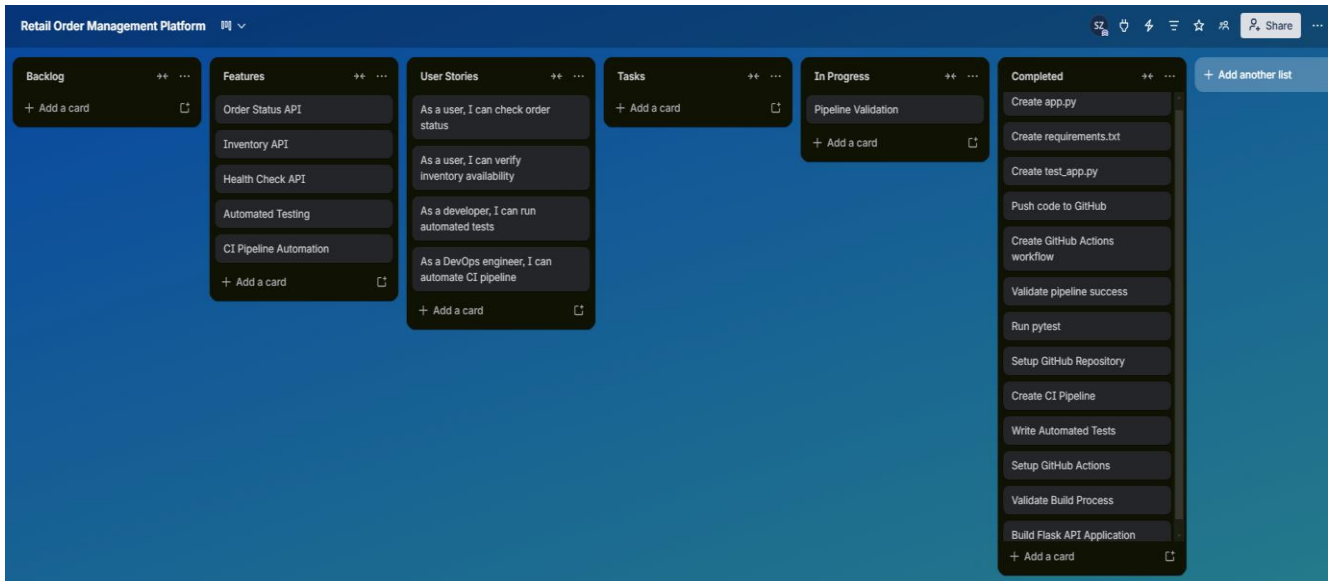
10. CI Pipeline Execution

The CI pipeline executed successfully in GitHub Actions and all automated tests passed successfully.

```
PS C:\Users\zerub\OneDrive\Desktop\retail-order-management-platform> git add .
PS C:\Users\zerub\OneDrive\Desktop\retail-order-management-platform> git commit -m "Added GitHub Actions CI pipeline"
[main 2331914] Added GitHub Actions CI pipeline
1 file changed, 27 insertions(+)
PS C:\Users\zerub\OneDrive\Desktop\retail-order-management-platform> git push
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 640 bytes | 71.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/ShinyZerubbaRajiveS/retail-order-management-platform.git
350d751..2331914 main -> main
PS C:\Users\zerub\OneDrive\Desktop\retail-order-management-platform>
```

The screenshot shows the GitHub Actions interface for the workflow 'python-app.yml' in the repository 'ShinyZerubbaRajiveS / retail-order-management-platform'. The workflow is titled 'Added GitHub Actions CI pipeline #3' and is in a 'Success' state. It was triggered by a push to the 'main' branch by user 'ShinyZerubbaRajiveS' at 2:33:19. The total duration of the workflow is 12 seconds. The workflow file is located at '.github/workflows/python-app.yml' and is triggered on 'push'. The workflow consists of a single job named 'build-and-test' which has a duration of 8 seconds. The interface includes a sidebar with navigation links for 'Summary', 'All jobs', 'build-and-test', 'Run details', 'Usage', and 'Workflow file'. The main area displays the workflow's execution details, including the trigger, status, duration, and artifacts.

11. Final Trello Board



12. Final GitHub Repo

 **retail-order-management-platform** Public Pin Watch 0

 **main**  **1 Branch**  **0 Tags** Add file Code

 **ShinyZerubbaRajiveS** Added GitHub Actions CI pipeline ✓ 2331914 · 9 minutes ago 3 Commits

 .github/workflows	Added GitHub Actions CI pipeline	9 minutes ago
 .pytest_cache	Added gitignore and removed pycache	14 minutes ago
 __pycache__	Initial retail order management application	19 minutes ago
 app.py	Initial retail order management application	19 minutes ago
 requirements.txt	Initial retail order management application	19 minutes ago
 test_app.py	Initial retail order management application	19 minutes ago